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Impacts of cartels on a market shown by the cartel case of car climate component suppliers

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1. Introduction

Competition requires firms to outperform similar companies in their market sector to be successful. Firms must either provide a better product than their competitor or make it possible to produce the same product with lower cost to obtain cheaper prices. This constant mechanism of competing with other firms results in continuously improving goods and services at lower prices for consumers and encourages innovation and efficient production within a company. (European Commission, 2015). Cartels impose a threat to those factors. Instead of optimising production or delivering better products, companies within a cartel agree to fix prices and limit production, thus consumers are provided with lower quality goods and services at non-competitive prices (European Commission, 2017). The goal of this paper is to analyse the negative impact of cartels by using economic tools, linking the analysis to the cartel case of climate component supplier, and to further determine the importance of fighting cartels.

Firstly, the paper provides a concise description of the cartel case. Secondly, the paper analyses the impacts cartels cause on the market due to economic inefficiency. Thirdly, each infringement of the case and their differences are described in more detail and linked to the economic analysis. Lastly, the paper summarises the economic importance of fighting cartels and ensuring competition within a market.

2. Description of the case

In March 2017, the European Commission (EC) fined the car climate control component suppliers Behr, Calsonic, Denso, Panasonic, Sanden and Valeo. Car climate control components include HVAC (heating, ventilation and air conditioning) units and engine cooling components. The companies were fined for illegal anti-competitive behaviour including coordinating prices and markets as well as exchanging sensitive information. At the end of 2010, Panasonic was the first company to apply for immunity under the leniency notice in respect to collusion with Denso regarding the supply of e-compressor, a component within a HVAC unit. Three further infringements were later uncovered by Denso's application for immunity in February 2011. Those infringements all took place between 2004 and 2009 and

were concerned with the sales of different components to numerous customers and varied in duration and parties involved. Each company cooperated with the EC and acknowledged their involvement during the settlement procedure, which made it possible to close the case in the beginning of 2017. Each company was fined its respective amount, while Panasonic and Denso were granted immunity for the infringements they revealed (European Commission, 2017).

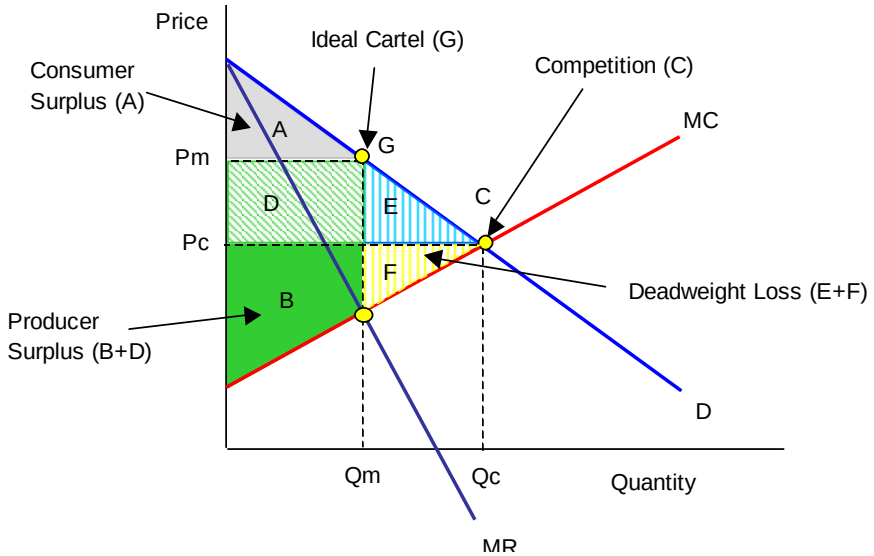
3. Economic analysis

The Commission tries to prevent and disrupt cartels as effectively as possible, because they alter the competitive market by deliberately limiting output to increase prices (European Commission, 2017). To determine why exactly the Commission wants to prevent this altering, this paper analyses three economic inefficiencies cartels can cause, summarised by Günster (2012). These three economic inefficiencies include: the allocative inefficiency and its impact on social welfare, the productive inefficiency within a firm, and the dynamic inefficiency that can result from cartels. According to Perloff (2018), a cartel acts the same as a monopoly with the difference of having an incentive to cheat. Out of simplicity reasons, the analysis ignores the incentive to cheat and underlies the assumption that cartels act like monopolies.

3.1 Allocative inefficiency

Allocative inefficiency is the status of a market where the price of a good does not equal the marginal cost, which means inefficient prices for consumers (Frantz, 2018). If companies act as a cartel, they gain market power and are able to become price-setters within a market, which means prices are going to differ from the competitive equilibrium and by that causes allocative inefficiency. The impact of this inefficiency can be seen at the net loss of society's welfare, which occurs in a non-competitive environment as Marshall (1890) argued. A common definition of society's total welfare consists of the consumer surplus and the producer surplus and takes both sides equally into account (Perloff, 2018). The consumer surplus is the amount a consumer is willing to pay for a certain quantity of a product minus the actual cost of this quantity (price). Producer surplus is the cost of a certain quantity of product (price) subtracted by the minimum amount a producer would be willing to produce this quantity (Perloff, 2018).

Figure 1. Loss in social welfare



Note. From "Cartels: Deadweight Loss and the Incentive to Cheat" by M. Sproul, Principles of Economics, p. 3. Copyright by California State University, Northridge

Those values respectively represent the well-being of each party and - as mentioned before are maximised in a competitive market. To see the net loss welfare experiences due to a cartel, the analysis refers to Figure 1. Following Perloff's (2018) definition of consumer and producer surplus, those are easily graphically identifiable in the figure. Under competitive conditions, meaning at price Pc and quantity Qc, the consumer surplus is represented by the areas A, D and E, while the areas B and F represent the producer surplus. Thus, total welfare consists of A, D, E, B and F (consumer surplus + producer surplus). The profit maximisation point for cartels, however is at MR=MC (Perloff, 2018). This causes the price to shift towards Pm and quantity towards Qm. This shift impacts the welfare of society in two different ways. First, area D is now part of the producer surplus instead of the consumer surplus. Total welfare is not reduced by that process, but a transfer of welfare occurs towards the producer side, which causes an unbalance in the market. Consumer welfare is reduced due to higher prices, while companies, which are part of a cartel, enjoy higher profit. Second, society's total welfare is reduced by the so-called deadweight loss. In the new market scenario with a cartel, society's welfare consists of A, B and D. Comparing this directly to society's welfare under competitive circumstances (A, D, E, B and F), cartels are not profitable for society' welfare, since they cause a net loss of E and F (deadweight loss) regardless of consumer or producer surplus. Thus, the European Commission fights cartels to firstly ensure that consumers enjoy fair and efficient prices and that secondly social welfare is maximised (European Commission, 2017).

3.2 Productive Inefficiency

The second type of economic inefficiency that can be caused due to cartels is productive inefficiency. A competitive company can only achieve profit maximisation if it produces efficiently, meaning using the smallest amount of input to produce its current output (Perloff, 2018). Companies acting in a cartel do not necessarily share that incentive. Leibenstein (1966) coined the term “x-inefficiency”, which describes the lack of motivation monopolistic or colluding companies have to produce efficiently (Huil, 2014). Leibenstein (1966) stated in his theory that monopolies, or in this case cartels, do not have an actual motivation to minimise cost. Since those companies have the market power to be price-setters in the market, they are able to set prices beneficial for them and not for the customer. Competition is the only force that incentivise companies to produce as efficiently as technologically possible, as they can then keep up with competing companies and sell products and services at lower prices. This only represents a non-allocative inefficiency, meaning an inefficiency within a company (Frantz, 2018), which does not necessarily harm the market or society itself, apart from the potential waste of input resources in the production.

3.3. Dynamic Inefficiency

The third type of economic inefficiency, dynamic inefficiency, is closely linked to the productive inefficiency within a firm and can harm the sector itself. Besides adopting new technologies, which colluding companies are already less inclined to, competitive firms may develop new technologies to stay ahead of competition and to further reduce production cost. The absence of the incentive to innovate is called dynamic inefficiency (Günster, 2011). Arrow (1962) described research as a “risky process” with an uncertain outcome to be economically beneficial in the future. Thus, incentives to invest in this “risky process” are reduced in non-competitive markets, meaning a market where either, knowledge is immediately shared, or monopolies/cartels dominate, because new technologies are not necessary or expected to increase profit. Günster’s (2012) empirical study confirms that research and development expenses of companies were lower during cartel years. Thus, the European Commission fights cartels, since they have a tendency to not invest resources to improve production methods and by that harm innovation and technological progress within their sector (European Commission, 2017).

4. Linking economic analysis to the case

Since the collusion was committed within the car component supply sector, it might be difficult to see how the different cartels affected the market. Nevertheless, we can use the economic assumptions of how cartels impact the market to develop an approximate picture. To do this successfully, the differences between the infringements are highlighted first. Those infringements are then linked to the assumptions of allocative inefficiency and dynamic inefficiency.

4.1 Differences of the four infringements

As mentioned in the description of the case, four separate infringements were identified, which vary in duration, companies involved, and customer affected. The first infringement included the companies Behr, Denso and Valeo, and concerned sales of HVAC components to multiple customers with focus on Volkswagen-group over a 4-year period. The second infringement included the companies Denso, Sanden and Valeo, and concerned sales of compressors to the customers Volkswagen-group and PAG over a 5-year period. The third infringement included Denso and Panasonic, who, for 5 months, agreed to share the supply of e-compressors towards Renault and Nissan. Finally, the forth infringement included Calsonic, Denso, Sanden and Valeo, and concerned the supply of HVAC and engine cooling components for specific car models of Suzuki, lasting for less than 2 years (European Commission, 2017).

4.2 Impact on social welfare

At first glance, it might seem that a cartel within the supply market will only affect the manufacturer itself. However, Perloff (2018) states that a non-competitive input market, such as cartels or monopolies, will ultimately cause the price of consumers to rise. In the stated case, the first two infringements included price coordination to avoid price reduction of the components (European Commission, 2017). This price coordination gave the suppliers market power i.e. opportunity to increase prices. Those higher input prices result in higher production cost for the manufacturers and eventually increase the price of the car as well (Perloff, 2018). Following the assumption of allocative inefficiency, those infringements did not only affect car manufacturers, but also the end consumer, whose social welfare was reduced due to higher, inefficient prices.

4.3 Impact on innovation

Dynamic inefficiency is a further consequence resulting from the lack of competition. The car component suppliers coordinated prices and markets, instead of competing and delivering better, innovative products. Vestager (2017), the European Commissioner, highlights the importance of each component involved in a car's production to achieve more efficient cars in the future. For example, according to Vestager's statement (2017), an e-compressor is a vital part of an electronic car and can immensely reduce energy consumption. Based on our assumption of dynamic inefficiency, the suppliers had a smaller incentive to innovate their product and discouraged technological progress within their sector during cartel years; this can heavily harm a future-orientated industry such as the electronic car industry, since technological stagnation is more likely to occur.

5. Conclusion

The European Commission (2017) argues that cartels create higher prices for lower-quality goods and services for consumers. To see if this holds true, this paper aimed to show the exact impact cartels have by analysing them from an economic standpoint and linking them to a real-life case. Therefore, this analysis has shown that cartels reduce society's welfare by a deadweight loss, but more importantly, reduce consumer welfare by increasing prices. In addition, cartels have a reduced incentive to produce efficiently and to innovate new technologies. The linking of the assumptions to the case further clarified how the lack of motivation to innovate can harm the whole sector and consumer, especially in industries that rely on technology. Thus, the enforcement of competition policies by institutions, such as the European Commission, is important. They ensure that consumers are protected, society's welfare is maximised and innovation within a sector is not hindered.

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